AI INFRA · FIELD NOTES

5 lessons from building a 19-agent AI system

the architecture, the decisions, and the trade-offs behind them

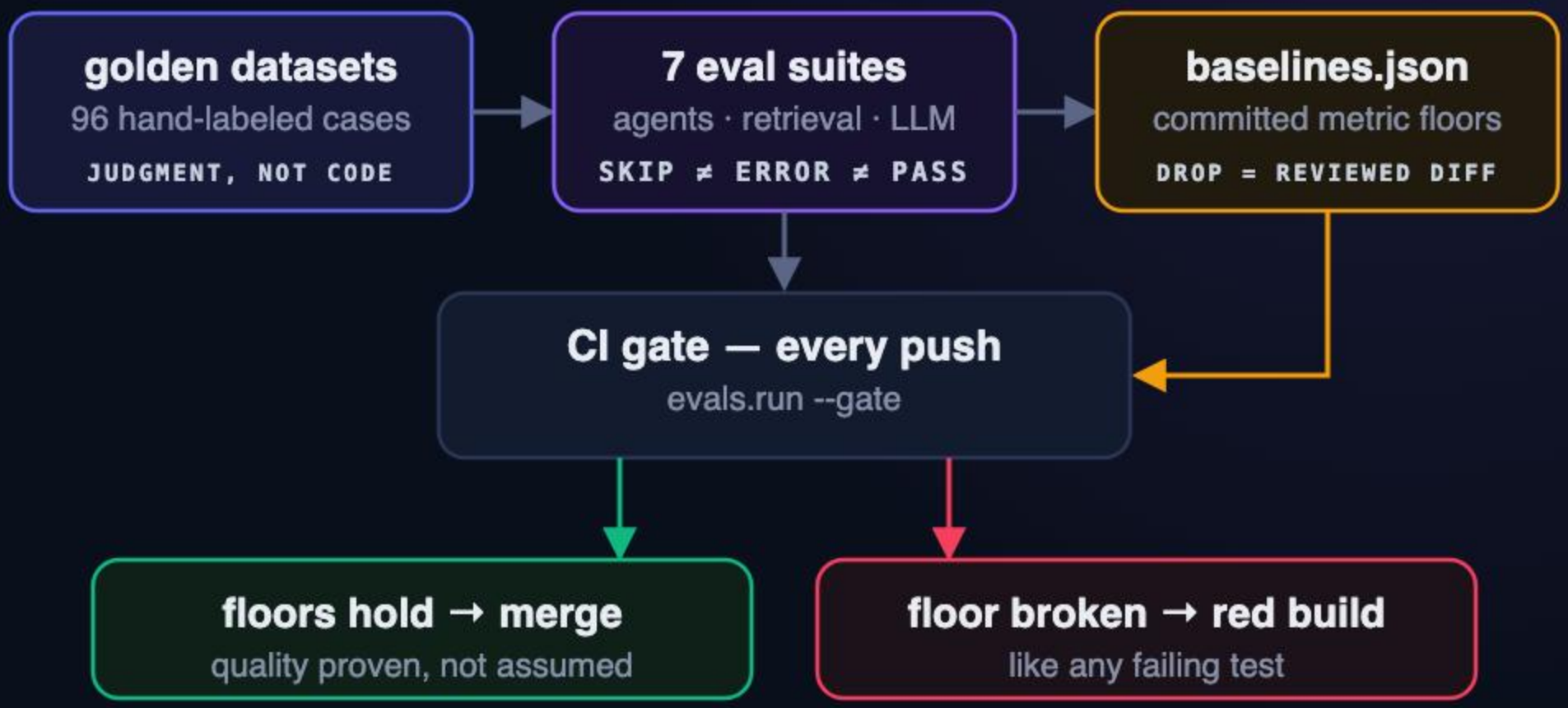


five stores, picked by access pattern — rows · ephemera · vectors · edges · blobs

> *Route work by latency budget: hot / warm / cold.*
> *Everything else in this system falls out of that decision.*

LESSON 01 · EVALS AS UNIT TESTS

Gate quality like code



first run: 3 real bugs found

`a \b cannot sit between a space and "$"`

→ money claims NEVER matched · fact-check recall was 0.73 · every test green

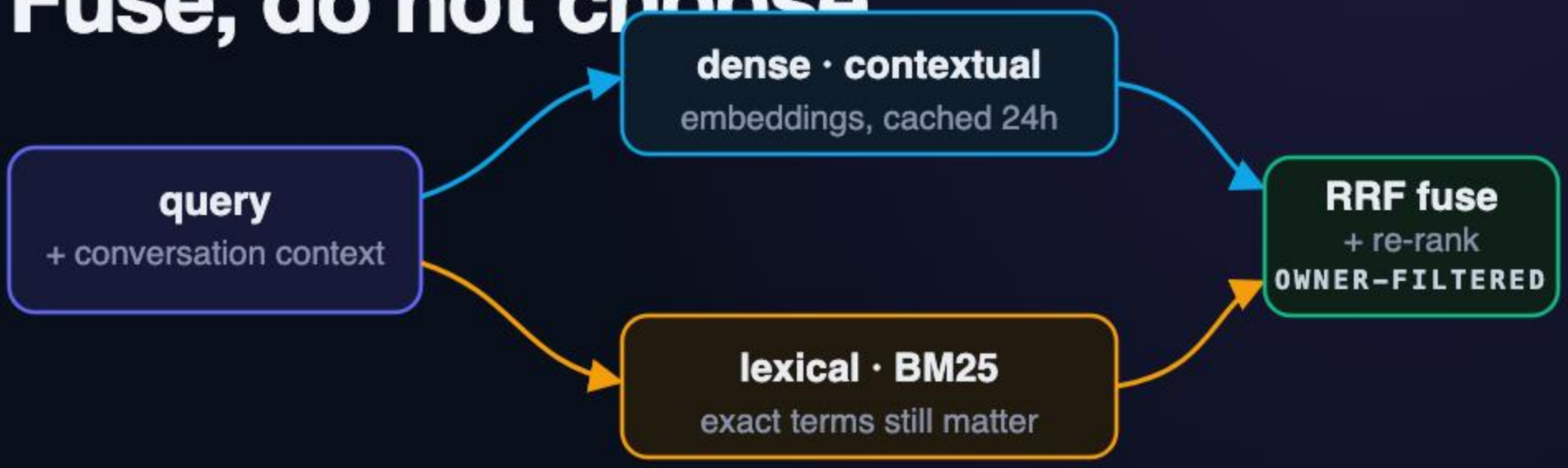
`fixed + floors ratcheted → recall 1.00, locked in CI`

⚖ THE TRADE-OFF

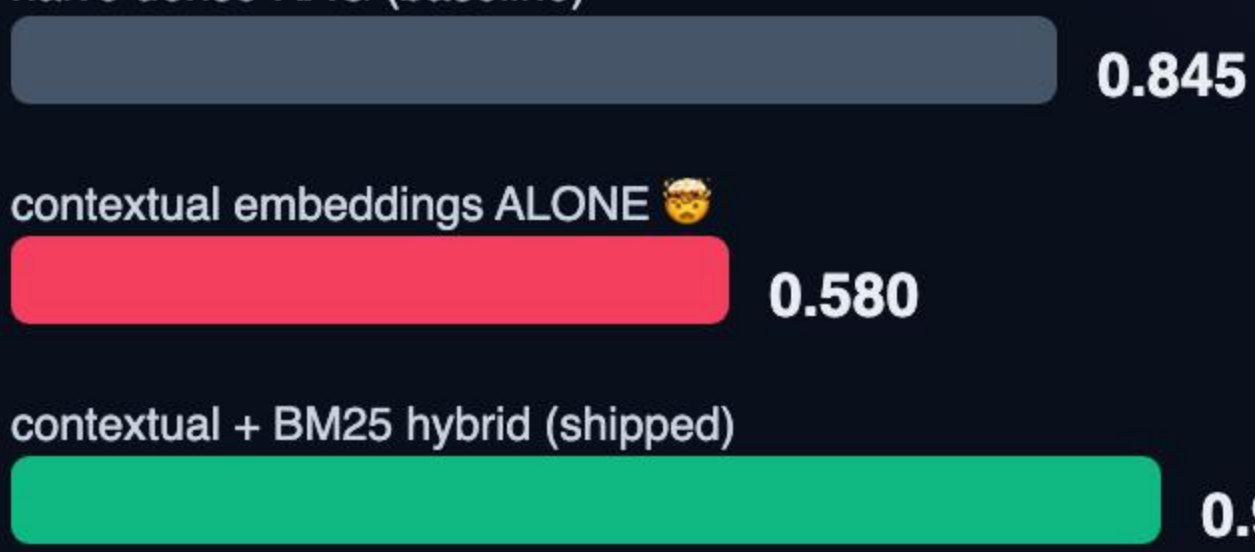
Hours hand-labeling golden cases — bought: prompt/regex changes can never silently make the product dumber again.

LESSON 02 · MEASURE RETRIEVAL

Fuse, do not choose



MRR · SAME MODEL, SAME QUERIES — ONLY THE STRATEGY DIFFERS



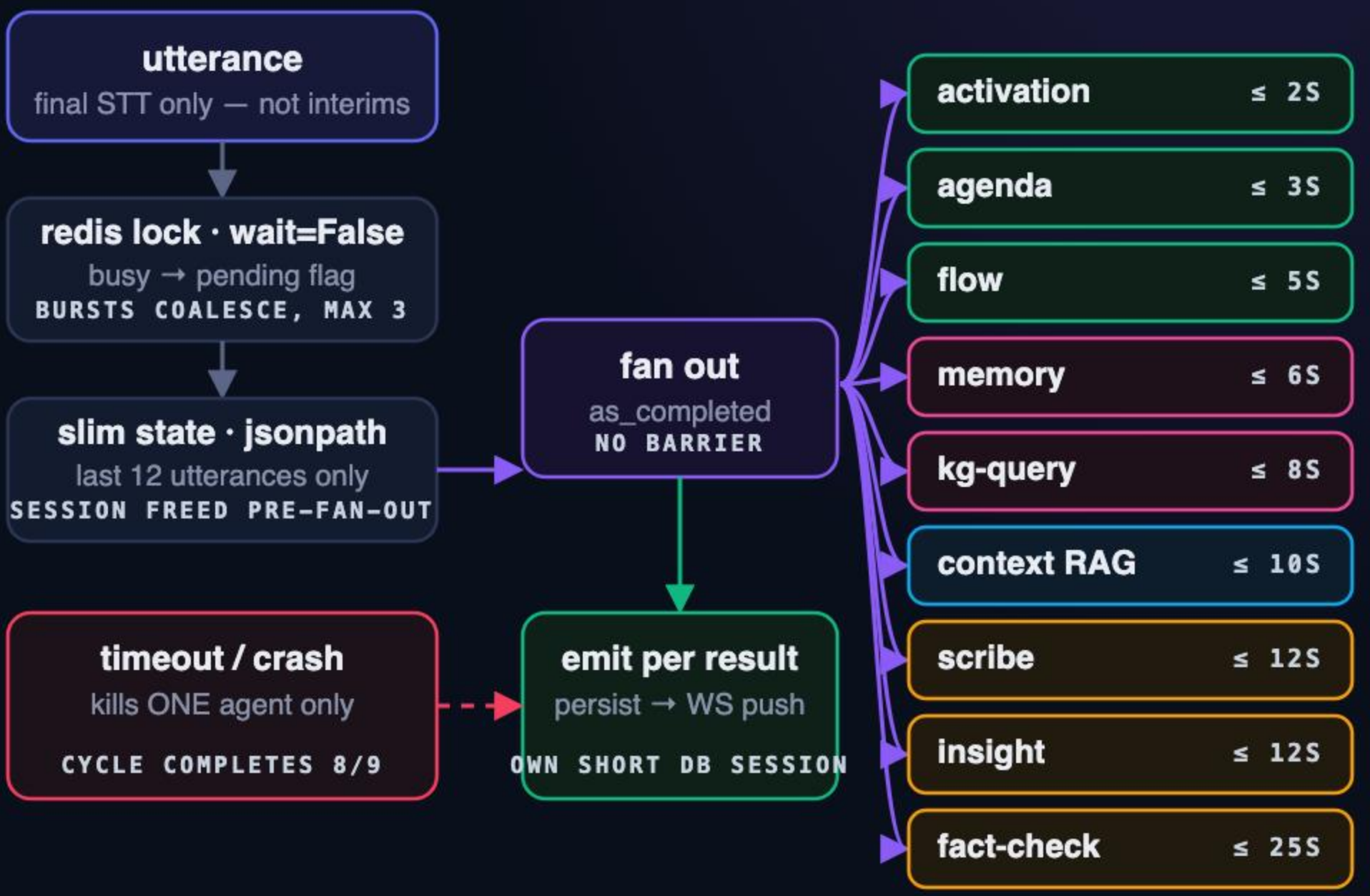
context enriches meaning — but homogenizes vectors within a document.
+9.9% MRR · +15.8% recall@1 over naive — and the lift is a CI gate now.

⚖ THE TRADE-OFF

Two index paths to maintain instead of one — bought: the fancy technique upside without shipping its measured downside.

LESSON 03 · STREAMING ORCHESTRATION

The hot path




```
gather(): first insight 20s
as_completed(): first insight 0.2s
```

⚖ THE TRADE-OFF

Atomic-batch simplicity traded for per-result emission + 9 timeout budgets — latency you can feel won over code you can shorten.

LESSON 04 · UNIT ECONOMICS

Typical utterance: $0

every utterance — all 9 agents look at it
100% of traffic

deterministic gates — regex · token match · si

most stop here · $0
6 of 9 live agents never call an LLM at all

only gated hits — structured extraction

gpt-4o-mini · ~15× cheaper
scribe · fact-check · prep · facts

only where prose IS the product

gpt-4o · 3 artifacts
synthesis · summary · answers

caches kill repeats
24h embeddings · owner-keyed answers

μUSD meter per meeting
GET /meetings/id/cost — pricing input

⚖ THE TRADE-OFF

Cheap gates could miss real decisions — so the gates themselves are eval-tested with recall floors. Cheap must never mean blind.

LESSON 05 · DESIGN FOR PARTIAL FAILURE

Degrade, do not die

OpenAI down	breaker trips → UI banner · meetings, transcripts, decisions keep working
Redis down	locks no-op · in-process fan-out · jobs fall back to BackgroundTasks
Neo4j down	graph features switch off · Postgres-only mode — enhancer, not dependency
one agent hangs	its timeout expires · the cycle completes with 8 of 9 results

…and the failures that never throw? Outcome-asserting tests catch what logs hide:

config that no-op'd
pydantic ignored env=
PROD MODE NEVER APPLIED

tenant-blind cache key
query-only answer cache
OWNER JOINED THE KEY

job that "completed"
worker hit localhost DB
ASSERT OUTCOMES, NOT LOGS

⚖ THE TRADE-OFF

Every fallback is more code and more tests — bought: no single vendor, store, or hung agent can take the meeting down.